PROJECT AND TEAM INFORMATION

Project Title

File Encryption & Decryption Tool

Student / Team Information

Team Number:	*T308*
Team member 1 (Team Lead)	*Rudransh Gupta –* *2510011500* *guptarudransh83@gmail.com* 
Team member 2	*Ritabrata Mandal –* *2510010593* *mandalritabrata50@gmail.com* 
Team member 3	*Nitesh Bishnoi –* *2510010972* *niteshkumarbishnoi933@gmail.com* 

PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

In our digital world, personal documents, educational records, source code, passwords, and a host of other sensitive information are often stored in files. If these files are sent or received without proper protection, it's all too easy for someone else to access that information. Plus, we can't ignore the issue of file size; larger files consume more storage space and bandwidth, which can lead to complications. That's why this project aimed to develop a File Encryption & Compression Tool to tackle these challenges head-on. This application allows users to choose a file, compress the data to save space, and then encrypt it to prevent unauthorized access. When permission is granted, the app can decrypt and decompress the files, granting access to the valuable data inside. Additionally, working on this project offers a fantastic opportunity to deepen our understanding of file operations, data structures, bitwise operators, algorithms, and the principles of object-oriented programming.

State of the Art / Current solution (1 pt)

Right now, we typically have two main types of software: one for encrypting files and another for compressing them. Encryption allows you to store files in a way that only someone with the right key or password can access, while compression helps shrink files down to a more manageable size. Since these tasks are usually handled by separate programs, you often end up needing to use more than one. Many of the available programs come packed with a bunch of complicated and often unnecessary features, which can be pretty overwhelming for those who are just getting familiar with these algorithms.

Project Goals and Milestones (2 pts)

Goals

- *Create a reliable tool that can compress and securely encrypt files.*
- *Set up operations for file input and output.*
- *Incorporate features for compression, decompression, encryption, and decryption.*

Milestones

- *Assess the requirements and design the overall system.*
- *Build and test the compression module.*
- *Develop and test the encryption and decryption functionalities.*

Project Approach (3 pts)

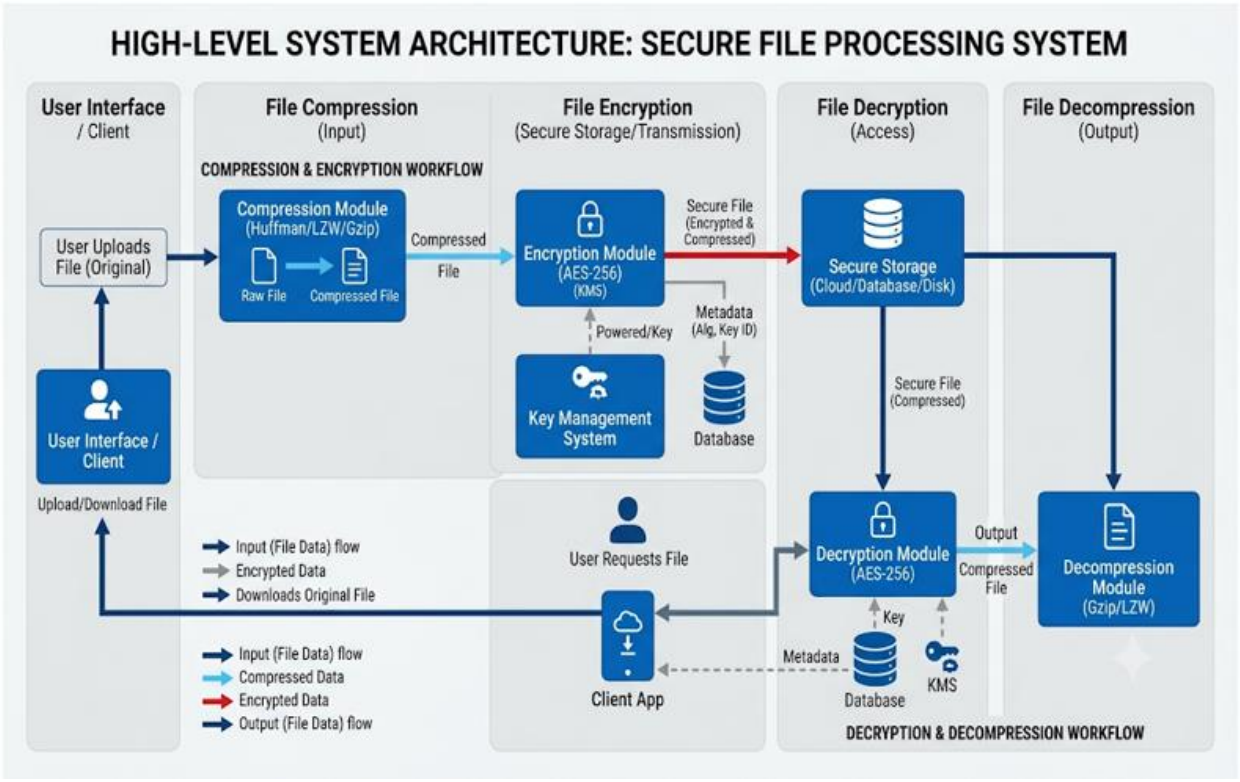
The application is crafted using C and C++, bringing together file handling, data structures, algorithms, bitwise operations, and object-oriented programming (OOP) to ensure files are securely protected. The journey of protecting a file kicks off with selecting the input file. From there, the application takes charge, compressing the data to shrink its size and encrypting it before saving it as a protected file.

File Protection: Input File → File Handling → Compression → Encryption → Protected File

File Recovery: Protected File → File Handling → Decryption → Decompression → Original File

The application is structured into various modules that handle file management, compression, encryption and decryption, as well as user interface management. C is primarily utilized for file handling and implementing fundamental data structures, while C++ takes the lead in organizing the application through object-oriented programming. We put the application to the test with a variety of file types to ensure that both encryption and decryption function smoothly. We also confirmed that compression and decompression maintain the integrity of the original file data. Additionally, we examined how the application deals with errors to guarantee that files can be processed reliably and recovered without any data loss.

System Architecture (High Level Diagram)



Project Outcome / Deliverables (1 pts)

The primary goal of this project is to create a user-friendly File Encryption and Compression Tool using C/C++. This application will empower users to easily select files and carry out tasks like encryption, decryption, compression, and decompression with just a few clicks.

Here's what the project will offer:

- *File Handling: Users can effortlessly choose files and manage all their input and output operations.*
- *Compression Module: This feature shrinks file sizes, helping to save valuable storage space.*
- *Encryption and Decryption Module: It secures files by encrypting them, while also allowing users to revert them back to their original state through decryption.*
- *Decompression and File Recovery: This functionality restores compressed files and assists users in retrieving their original data.*

Assumptions

The program relies on input files that users can easily access and read. It's primarily intended for educational use, showcasing the principles of encryption and compression, rather than serving as a substitute for professional security or compression software. You'll also need a C/C++ compiler and the appropriate development environment to get started. If you're the rightful owner, you can recover the original files by using the correct encryption key or password.